

SafeRoad India

Interactive Road Accident Analytics Dashboard
Data Analytics Project using Indian Road Accidents Dataset
(2022–2025)

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Problem Statement



Why Road Accident Analysis?

- Road accidents continue to be one of the major causes of injuries and deaths throughout India.
- The public domain contains a lot of accident data, but it is hard to make sense of raw datasets without the right tools for analysis and visualization.



Project Overview



What is SafeRoadIndia?

SafeRoadIndia is an interactive web-based dashboard created using Python & Streamlit to evaluate Indian road accident statistics from 2022 - 2025.

The dashboard translates raw accident statistics into relevant visualizations so that users may obtain an idea of geographical distribution, accident severity, temporal patterns, and contributing factors using interactive filters and analytical charts.



Objectives



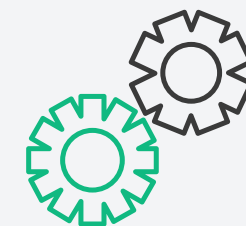
**Analyze accident trends
across India.**



**Identify accident-
prone states and
cities.**



**Study weather, road,
and cause related
accident patterns.**



**Provide an
interactive and
user-friendly
analytics platform.**



Kaggle – Indian Road Accident Dataset(2022-2025)

- 20,000 Accident Records
- Multiple States
- Multiple Cities

- Year
- State
- City
- Latitude & Longitude
- Weather
- Road Type
- Accident Severity
- Cause
- Day of Week
- Hour
- Time of Day

ID	City	State	Latitude	Longitude	Date	Time	Hot Day
0	Pune	Maharashtra	18.68	73.93	2023-10-20	05:00	Sun
1	Mumbai	Maharashtra	18.81	72.79	2023-05-20	04:00	Sun
2	Mumbai	Maharashtra	19.09	72.81	2024-07-11	13:00	Wed
3	Chandigarh	Punjab	30.78	76.84	2025-03-23	11:00	Sun
4	Chennai	Tamil Nadu	12.96	80.28	2024-01-21	16:00	Thu
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Dashboard Features

Filters & ML Risk prediction Sidebar Key Performance Indicators (KPIs) & Visualizations

 **Filters**

Select Year

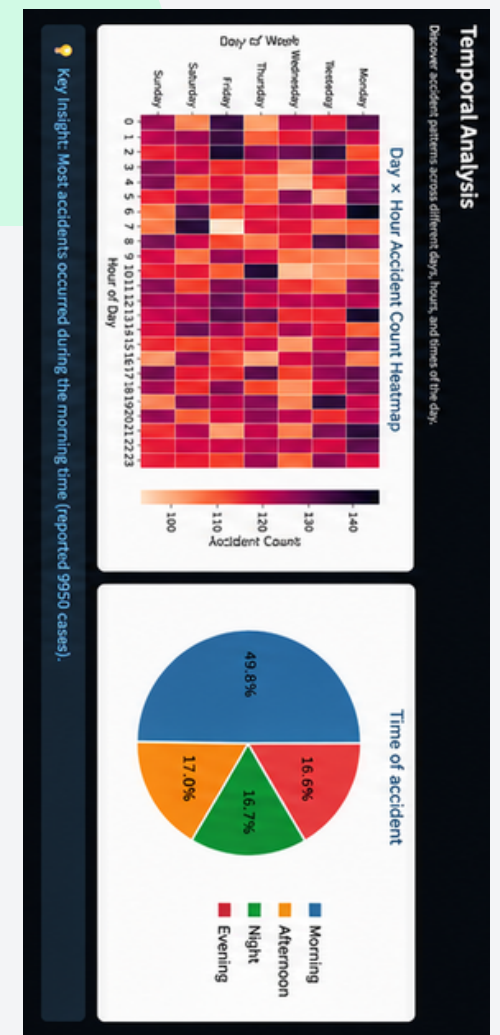
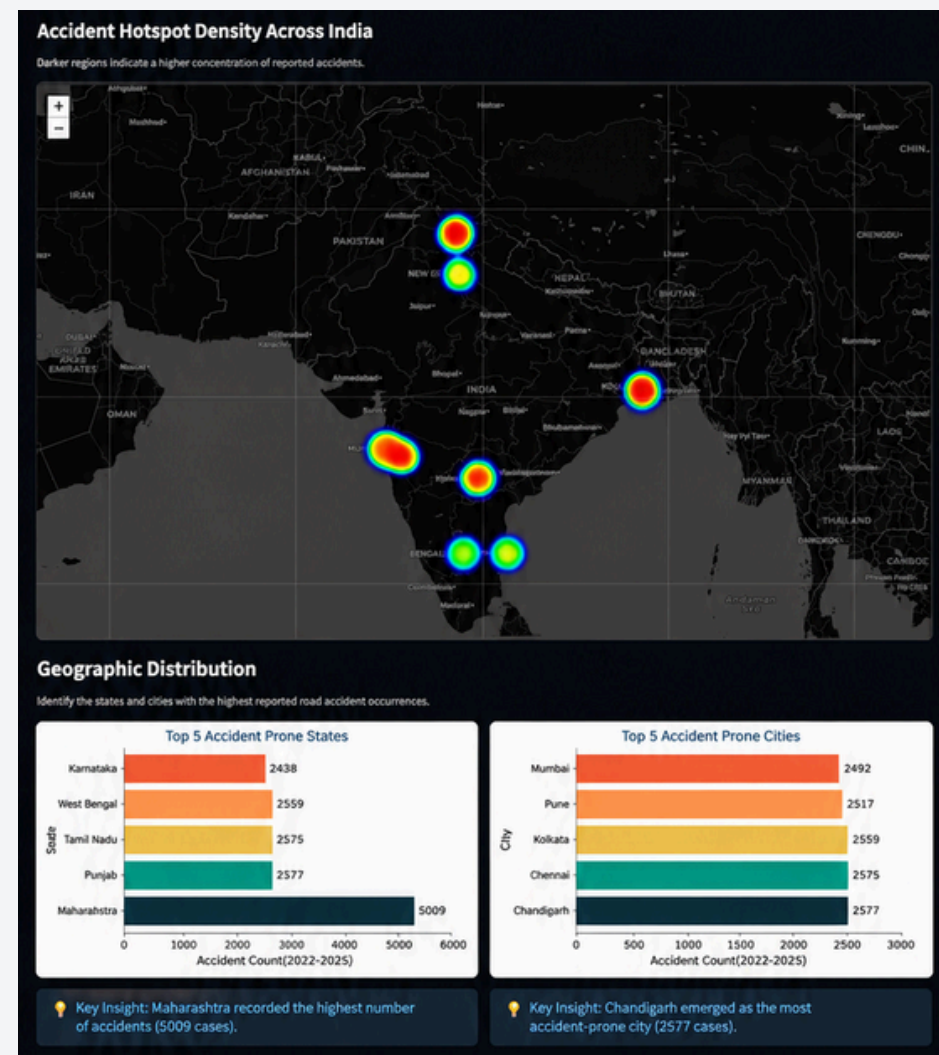
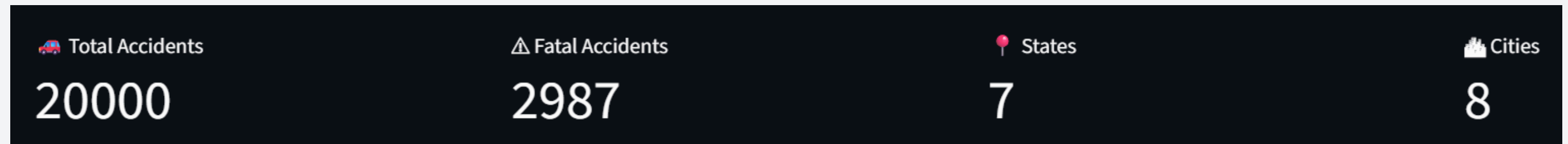
All

Select State

All

Select City

All





Conclusion

- Successfully transformed accident data into meaningful insights.
- Dashboard improves understanding of accident patterns.
- Interactive visualization simplifies exploration.

Thank you

Next up, Streamlit dashboard:

